

Harsh Gupta

Data Engineer — Data Scientist — Machine Learning & Deep Learning

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PROFESSIONAL SUMMARY

Data Engineer and ML practitioner experienced in ETL/ELT pipelines, star-schema modeling, and automation bots for quick-commerce scraping. Skilled in end-to-end ML, RAG, Transformers, and predictive modeling (Kaggle Top 10%). Turns raw data into actionable insights and production systems.

TECHNICAL SKILLS

Languages	Python, C++, SQL, C, Bash
Data Eng.	ETL/ELT, Star Schema, Airflow (basics), DBT (familiar), Data Modeling, Web Scraping (Selenium, Playwright)
ML/DL	PyTorch, TensorFlow/Keras, Scikit-learn, XGBoost, OpenCV, MONAI
LLMs & NLP	Hugging Face Transformers, LangChain, LlamaIndex, RAG, BERT, GPT, Groq
Analytics	Pandas, NumPy, Matplotlib, Seaborn, Tableau, Feature Engineering
MLOps & Infra	Docker, Git, MLflow, FastAPI, AWS (EC2, S3), Linux, CI/CD, Prometheus, Grafana

EXPERIENCE

Data Engineer Intern May 2026 – Present
Visage – Bombay Shaving Company / 100 Days Gurgaon, India

- Designed ETL/ELT pipelines (Python, SQL, orchestration) to ingest commerce data for Durex, Colgate, Bikaji; built star-schema models for real-time P&L dashboards.
- Developed automation bots (Selenium, Playwright) scraping Blinkit/Zepto pricing, availability, and competitor data, feeding central analytics.
- Integrated Prometheus & Grafana for data freshness monitoring, reducing incident detection time by 40%.

Research Intern May 2025 – Jul 2025
Defence Research & Development Organisation (DRDO), DEAL Dehradun, India

- Engineered embedded C firmware for a ground-based PSoC system to establish a robust wireless link with an airborne hovercraft; applied digital filtering that cut signal noise by 18%, enabling reliable real-time telemetry.
- Built Python dashboards that visualized live aircraft telemetry and flagged anomalies, directly supporting test team decisions.

Robotics Lead Jul 2023 – Dec 2023
Autonomous Systems Team, Garuda Aerospace Indore, India

- Developed a custom ResNet-18 for agricultural analytics, boosting disease detection accuracy by 12% using aerial imaging.

EDUCATION

2023 – 2027 B.Tech, Electronics & Communication, **IIT Bhopal** CGPA: 8.48/10
2024 – Present Minor in Data Science, **IIT Madras** (Online)

Relevant Coursework: Machine Learning, Deep Learning, NLP, Probability, Linear Algebra, Data Structures

RESEARCH & PUBLICATIONS

- Physics-Constrained ResNet for Brain Tumor Segmentation & Glioma Growth Modeling** *In Preparation*
Developed a 3D MONAI ResNet with Fisher-KPP reaction-diffusion PDE constraints for joint segmentation and biophysical parameter estimation on BraTS.

PROJECTS

- YouTube AI Assistant (Chrome Extension)** 
Context-aware extension with multi-model fallback (Gemini, DeepSeek, Groq) and RAG over video transcripts. Includes code explanation, Wikipedia integration, and semantic search.
- AI Stock Analyst Telegram Bot** 
Serverless agent delivering real-time technical analysis, news sentiment, and fundamentals via Telegram. Multi-API orchestration with optimized token usage.
- Luffy Music Player — AI Recommendation & Playlist Storage**  
Full-stack web app combining content-based (KNN on 14+ audio features) and collaborative filtering (SVD) for personalised music recommendations. Features infinite scroll, mood filters, command palette, YouTube integration, and a One Piece-themed UI; served via FastAPI and deployed on Render.

CERTIFICATIONS

- May 2024** — Deep Learning Specialization (deeplearning.ai) — 5 courses
- Mar 2024** — Hugging Face NLP Course — Transformers, fine-tuning

ACHIEVEMENTS

- 1st Place** — Smart India Hackathon 2024 (Hardware Edition) — AI-powered agricultural drone.
- Top 10%** — Kaggle "House Prices" (0.121 RMSLE); Top 12% — "NLP with Disaster Tweets".
- ICPC Regionalist 2024** — Rank 412 among 3000+ teams in India.
- KVPY Scholar 2021** — Dept. of Science & Technology, Govt. of India.
- Codeforces Expert** — Rating **1747** (max. Expert, **1788**); 900+ problems solved.